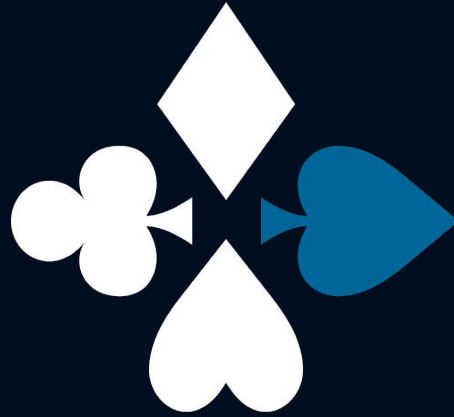




Pokerbots 2026

Lecture 1: Introduction to Pokerbots

Meet the team



Sri Saraf
Co-President



Bobby Costin
Co-President



Sejal Rathi
Vice President

Meet the team



Jacob David
Head of Curriculum



Jacob Jobraeel
Innovation Chair

Meet the team



Paco Gomez-Paz
Engine Lead

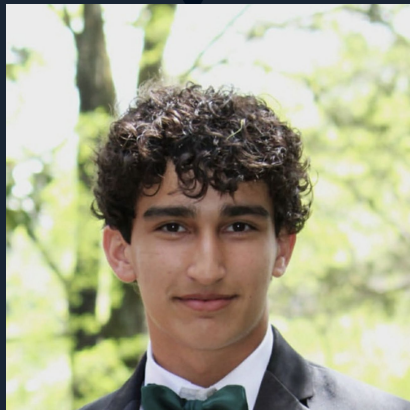


Steve Zhang
Server Lead



Sam Zhou
Tech Lead

Meet the team



Landon Sims
Engine Developer



Reese Long
Server Developer



Alan Duan
Treasurer



Michelle Gao
Sponsorship Director

Sponsors



CITADEL | CITADEL Securities



Register & Resume Drop:
pkr.bot/reg

Giveaways!

Giveaway Game:

pkr.bot/count



- Guess the number of people who will be in this room after lunch break
- You can be as precise as you want, but enter a decimal number (eg. 100.6)
- Winner is who guesses the closest (and earliest if there's a tie)
- Prize: Sansui Curved 24 Inch Gaming Monitor 160 Hz

Random Raffle:

pkr.bot/random

Prize:

- Beats Studio Pro



Agenda

- Class overview + logistical details
- Intro to poker + variant rules
- Skeleton bot setup and connecting to the scrimmage server
- Testing your bot locally
- Overview of skeleton bot architecture
- Live Coding Lecture 1 reference bot

Class Overview + Logistical Details

Resources

- Github: pkr.bot/resources
 - Updated Syllabus (Schedule and all logistical info is here)
 - (syllabus will be renewed when schedule is finalized)
 - Lecture slides, variant info, game rules
- Piazza: pkr.bot/piazza
 - All updates will be posted here
 - Best place to ask questions to staff or classmates

Class Structure

- ~Twelve lectures
 - MTWR 12:00-1:30pm EST, 6-120
- Recitation and Office hours
 - MTWR 2-4pm EST, 26-168
 - Office Hours virtual on days with no recitation
- Scrimmage server
- Weekly tournaments
- Final tournament and event (Jan 30th)

Team Formation

- Teams are 1-4 people
- Google Sheet: pkr.bot/teammates
- Poker Afternoon Study Breaks!
 - This Wednesday during recitation
 - Play Poker and try out the 2026 Variant!
 - Great place to find teammates
- Team Specific Prizes



>\$50K

Total Prize Amount

Prizes

- Grand prize: \$10k
- 2nd - 5th place : \$6.5k, \$3.5k, \$2k, \$1k
- Bot Language Specific Prizes
- ~\$2.5k/week Scrimmage Server Prizes
- All announced prizes can be viewed on the syllabus
- More prizes to be announced soon!

Lecture Series

- Introduction to Pokerbots
- Probability and Statistics
- Poker Strategy
- Machine Learning
- Game Theory
- Reinforcement Learning
- Neural Networks
- Counterfactual Regret Minimization
- Decision Trees
- ~And More Surprise Lectures~

How to Receive Credit For This Course

Participate

Submit a bot to the
scrimmage server
every week

Improve

Each week you
should beat your
bot from the
previous week

Report

Submit a 3-5 page
strategy report at
the end of the
competition

Questions?



Intro to Poker + Variant Rules

Heads-up No-limit Hold'em Poker Rules

- Objective: earn as many chips as possible
- Players receive two secret cards (*hole cards*)
- Players bet into a pot in several rounds; the pot is given to the player with the best poker hand
- Structure of a betting round: first player can bet 0 (*check*) or any amount between “big blind” and the # of chips they have remaining. If they check, action passes to second player. If they bet, second player can *fold*, *call*, or *raise*
- Hand strength is determined as the best 5-card poker hand you can form out of 7 cards: 2 hole cards and 5 community cards

Poker Rules

- Two cards dealt to each player (*hole cards*)
- Betting round (first betting round begins with *blinds*)
- Flop (3 community cards revealed)
- Betting round
- Turn (+1 community cards revealed [4 total])
- Betting round
- River (+1 community cards revealed [5 total])
- Betting round
- Settlement (*showdown*)

Card Rankings

THE 13 RANKS					
	A	K	Q	J	
	ACE	KING	QUEEN	JACK	
10	9	8	7	6	5
10	9	8	7	6	5
4	3	2			
4	3	2			

THE FOUR SUITS			
3	3	3	3
CLUBS	HEARTS	SPADES	DIAMONDS

Poker Hand Strength

Hand Rankings

ROYAL FLUSH



STRAIGHT FLUSH



FOUR OF A KIND



FULL HOUSE



FLUSH



STRAIGHT



THREE OF A KIND



TWO PAIR



ONE PAIR



Pokerbots Game Iteration

- 1000 rounds
- 400 chip stack, resets each round
- Small/big blinds: 1/2
- Chip profits/losses from all the rounds are totaled to determine winner of match



2026 Variant Reveal

2026 Variant - Toss or Hold'em

Both players are dealt three hole cards instead of two:

- The flop is then two cards which can be seen by both players
- The players sequentially toss one of their cards into the board
- The board then contains 4 cards, followed by a turn (5 cards) and a river (6 cards)
- Best 5-card hand wins

GOAL:

Take advantage of your toss choice while trying to deduce your opponent's info!

Detailed rules for the variant will be posted
on the class resources page after lecture

Skeleton Bot Setup: Connecting to the Scrimmage Server

Before You Code

Info on Piazza about development environment for Pokerbots

Once finished, make a GitHub account to facilitate code management

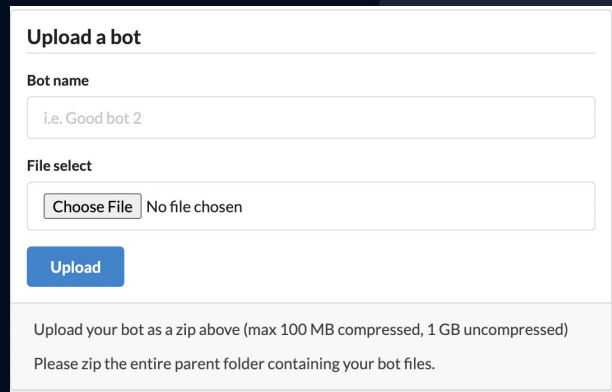
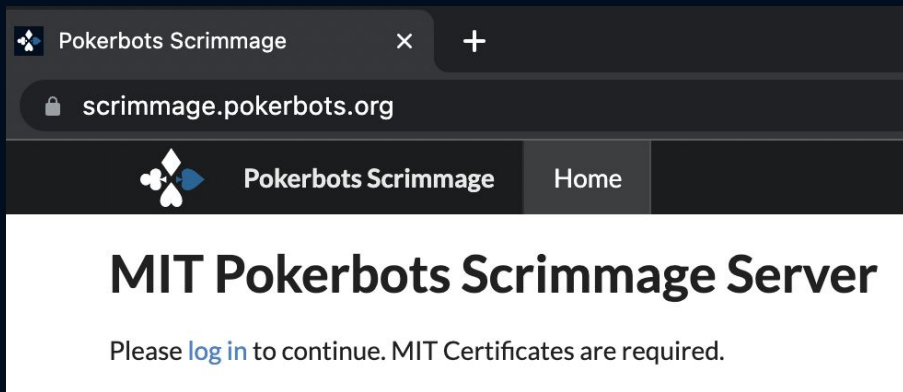
Read up on using the command line and Git

We can help with these steps during recitation/office hours after lecture

Getting Set Up

Clone engine from mitpokerbots/engine-2026; create new repository for your code

Scrimmage server: download ZIP file and upload it directly.

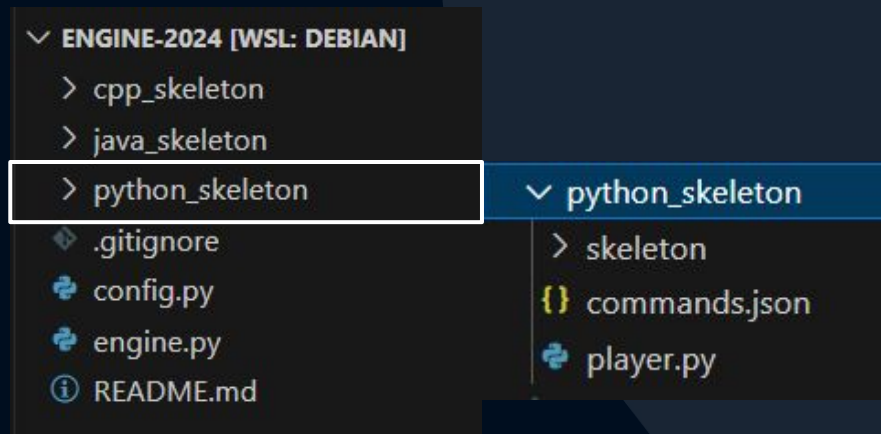




Engine Structure

Engine Structure

- 3 Skeleton directories
 - One for each language
 - Edit code in player.py, Player.java, main.cpp
 - Zip this for the scrimmage server
- Player Chatbot
 - Easy to play against your bots
- engine.py
 - Core engine instructions
 - Run this to test a game
- config.py
 - Parameters for game





Overview of Skeleton Bot Architecture

Skeleton Bot Structure

- GameState, RoundState classes
- handle_new_round(), handle_round_over(), get_action()
- Skeleton Bot walkthrough

Coding reference-1-2026 bot

Example Bot

- We want to analyze our hole cards
- We want to toss the “worst” one
- Eventual goal: use a more advanced method to place a bid combined with more sophisticated betting logic

Lunch 🤪

- To get lunch, just show proof that you submitted your resume at pkr.bot/reg



Domino's®



Live Coding Session

All code from today can be found at:

github.com/mitpokerbots/reference-1-2026

Note: not much code was changed, mostly just showing how to set up your environment

Giveaway Winners

Count Guessing Game

Winner: kerb “roshanp”



With a guess of 36!

Raffle Winner: kerb “djzhang”



Thanks for watching!

Slides/notes will be posted on pkr.bot/resources

Make sure to check pkr.bot/piazza for updates

Lecture recordings at pkr.bot/panopto

Leave feedback at pkr.bot/feedback!